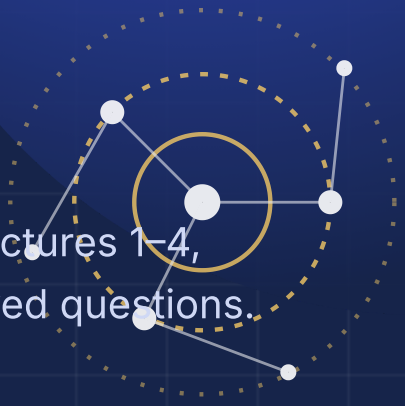
CSE557 · BIG DATA ANALYTICS · 5TH SEM

Unit 1
Study Guide



From zero to exam-ready: every slide from Lectures 1–4, explained simply, with diagrams, tips and solved questions.

1 **Types of Digital Data**
Structured · Semi-structured · Unstructured · ACID · Text mining

2 **Introduction to Big Data**
Evolution · Definition · 3 Vs · Challenges · BI vs Big Data

3 **Big Data Analytics**
Analytics 1.0/2.0/3.0 · Data Science · Data Scientist · SMP/MPP · CAP

Textbook: Seema Acharya & Subhashini Chellappan, *Big Data and Analytics* (Wiley) CIE-1 Edition

How to use this guide

This guide assumes you know **nothing** about Big Data yet. Read it in order: each chapter builds on the previous one. Everything from your Unit 1 slides (Lectures 1–4) is here, rewritten in simple language, with the textbook terms you need to write in the exam.

What the coloured boxes mean

🔑 **KEY POINT**
Must-know fact. If you remember nothing else, remember these.

🎯 **EXAM TIP**
How this topic is usually asked and how to score full marks.

💡 **TIP**
A shortcut to understand or study the idea faster.

🧠 **MEMORY TRICK**
Mnemonics to recall lists quickly in the exam hall.

⚠️ **COMMON MISTAKE**
Things students mix up — avoid losing marks here.

📌 **EXAMPLE**
A real-life example that makes the concept click.

Also: yellow highlighted text = exam keywords. Use these exact words in your answers.

Suggested study plan (about 6–7 hours total)

Session 1
Chapter 1 — Types of Digital Data
≈ 1.5 hrs

Session 2
Chapter 2 — Introduction to Big Data
≈ 1.5 hrs

Session 3
Chapter 3 Part A — Analytics, Data Science
≈ 1.5 hrs

Session 4
Ch 3 Part B — Terminologies (SMP, MPP, CAP) + Cheat sheet
≈ 1.5 hrs

💡 **HOW TO STUDY EACH SECTION**

1. **Read the "In simple words" box first** — get the idea before the definition.
2. **Look at the diagram** and try to explain it out loud in 2–3 lines.
3. **Learn the textbook definition** (the boxed *Definition*) — examiners like the exact wording.
4. At the end of every chapter, tick the **"Can you...?" checklist**. If you can't tick something, go back.
5. The night before the exam, read only the **Cheat Sheet** (last pages) and the 🔑 boxes.

🎯 **ABOUT CIE-1**

CIE-1 is for **30 marks** and covers CO1, CO2 & CO3 → **Unit 1, Unit 2 and the first part of Unit 3** (Spark architecture overview). This guide covers **Unit 1 completely**. For long answers (5–10 marks), always write: **definition → explanation in points → diagram → example**. A neat diagram is often worth 1–2 marks by itself.

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CHAPTER 1

Types of Digital Data

What you'll learn

1. What data is and where it comes from
2. The 3 types of digital data
3. Structured data, RDBMS & ACID
4. Semi-structured data: XML & JSON
5. Unstructured data & its sources
6. Techniques to deal with unstructured data
7. The big comparison table
8. Solved sample questions

⏱ ≈ 1.5 hours | Slides: Lecture 1 (pp. 8–46) | Textbook Ch. 1

1.1 What is data?

🗣 IN SIMPLE WORDS

Data = **raw facts**. Numbers, words, pictures, sounds — anything a computer can store and work on. On its own data may not mean much; once we organise it, it becomes **information**, and when we understand it, it gives **insights**.

DEFINITION

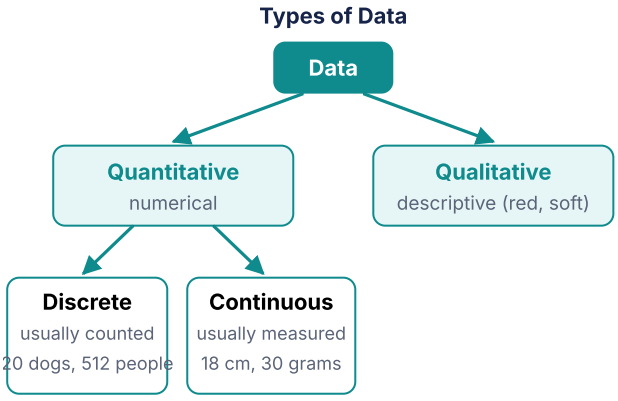
Data is the quantities, characters, or symbols on which operations are performed by a computer, which may be **stored and transmitted** in the form of **electrical signals** and recorded on **magnetic, optical, or mechanical** recording media.

Data comes from...

Data can be numbers | facts | quantities | information | graphs | observations | measurements

Computer data as information

- Computer data is **information processed or stored by a computer**.
- It may be text documents, images, audio clips, software programs, or other types of data.



- It is processed by the computer's **CPU** and stored in **files and folders** on the hard disk.

Where does data live?

- Data is present internal to the enterprise and also exists outside the four walls and firewalls of the enterprise.
- Data is present in **homogeneous** sources (all of the same kind) as well as in **heterogeneous** sources (of many different kinds).

Data raw facts → **Information** organised, meaningful data → **Insights** understanding that helps decide

1.2 Classification of digital data

IN SIMPLE WORDS

Think of three cupboards. **Structured** = everything in labelled boxes in neat rows (an Excel sheet). **Semi-structured** = items with name-tags but no fixed boxes (an XML/JSON file, an email header). **Unstructured** = everything thrown in a pile (photos, videos, chats).

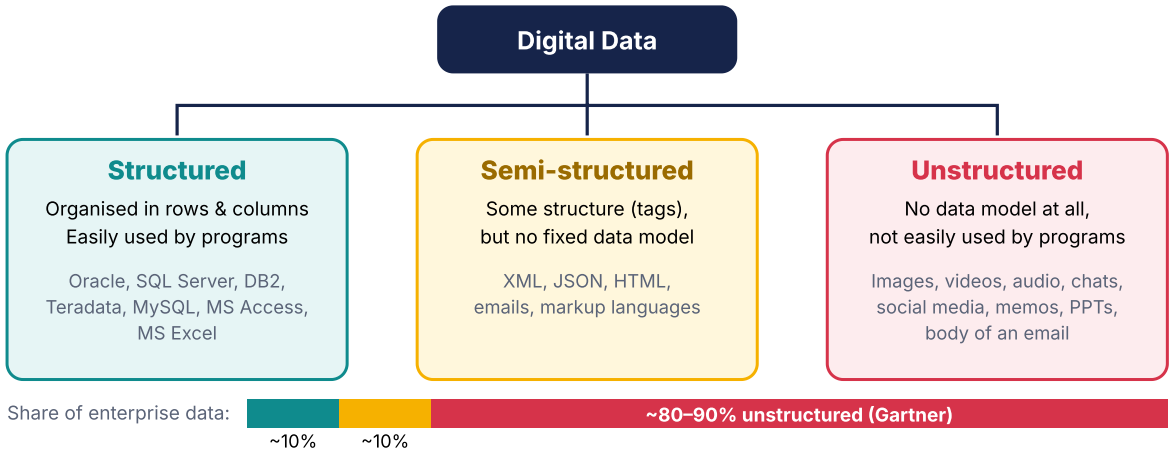


Figure 1.1 — The three types of digital data

KEY POINT

About 80–90% of an organisation's data is unstructured. Only a small part is neatly structured. This is the main reason we need Big Data tools.

1.3 Structured data

DEFINITION

Structured data is data which is in an organised form (e.g., in rows and columns) and can be easily used by a computer program. Relationships exist between entities of data, such as classes and their objects.

Example: data stored in databases.

Structured data and RDBMS

- Most structured data is held in an **RDBMS** (Relational Database Management System — e.g., Oracle, MySQL, SQL Server, DB2).
- An RDBMS follows the **relational data model**: data is stored in **rows/columns** (a table is also called a *relation*).
- The number of rows/records/tuples in a relation = cardinality of the relation.
- The number of columns = degree of the relation.

🧠 **MEMORY TRICK**

Cardinality = **C**ount of rows (records). **D**egree = **D**own-the-top columns (attributes). Rows go "card by card", columns give the table its "degree" of detail.

How do we design structured data? (steps)

1. **Design the relation/table** — decide the fields/columns and the **type** of data each will store: number (integer or real), alphabets, date, Boolean, etc.
2. **Decide the constraints** the data must follow:
   - **UNIQUE** values in a column, **NOT NULL** values in a column,
   - a **business constraint** e.g., value in a column should not drop below 50,
   - a **set of permissible values** e.g., the column should accept only "CS", "IS", "MS".
3. **Relate tables** to each other using common columns.

📌 EXAMPLE — EMPLOYEE TABLE (FROM YOUR SLIDES)

Column	Data type	Constraint
EmpNo	Varchar(10)	**PRIMARY KEY**
EmpName	Varchar(50)	
Designation	Varchar(25)	NOT NULL
DeptNo	Varchar(5)	
ContactNo	Varchar(10)	NOT NULL

Schema of the "Employee" table

EmpNo	Name	Designation	Dept	ContactNo
E101	Allen	Software Engineer	D1	0999999999
E102	Simon	Consultant	D1	0777777777

Sample records → cardinality = 2, degree = 5



Figure 1.2 — Tables in an RDBMS can be related through a common column

Sources of structured data

Databases such as Oracle, IBM DB2, Microsoft SQL Server, Teradata, MySQL, PostgreSQL, plus MS Access, MS Excel (spreadsheets) and OLTP (online transaction processing) systems.

Ease of working with structured data

Structured data is the easiest type to work with. Your slides list **5 reasons**:

✏️ 1. Insert / Update / Delete

The **DML (Data Manipulation Language)** operations make it easy to input, store, access, process and analyse data.

🔒 2. Security

Strong **encryption and tokenization** protect data throughout its lifecycle. Only **authorised** people can decrypt and view sensitive data → organisations stay in control and compliant.

🔎 3. Indexing / Searching

An **index** is a data structure that **speeds up data retrieval** (mainly SELECT queries), at the cost of extra writes and storage space — but faster search is worth it.

📈 4. Scalability

Storage and processing capacity of an RDBMS can be scaled up by **increasing the horsepower of the server** (more processor power, primary & secondary storage).

🔁 5. Transaction processing — ACID properties

RDBMS supports the **ACID** properties of a transaction (a transaction = a unit of work, e.g., transferring money):

A · Atomicity
All or nothing — a transaction happens **in its entirety or not at all**.

C · Consistency
DB moves from one consistent state to another. Same data stored in 2+ places must **agree**.

I · Isolation
Each transaction runs as if it were the **only transaction** happening.

D · Durability
Once done, changes are **permanent** (survive crashes).

🧠 **MEMORY TRICK — 5 EASES OF STRUCTURED DATA**

"I S I S T" → **I**nsert/update/delete · **S**ecurity · **I**ndexing · **S**calability · **T**ransaction processing (ACID).

📌 **ACID WITH A BANK EXAMPLE**

You send ₹500 to a friend. **Atomicity**: money leaves your account *and* reaches theirs — never only one. **Consistency**: total money in the bank is still correct. **Isolation**: another transfer at the same moment doesn't mix with yours. **Durability**: after "Success", a power cut won't undo it.

1.4 Semi-structured data

DEFINITION

Semi-structured data is data which does not conform to a data model but has some structure. It is not in a form which can be used easily by a computer program. It is also called a self-describing structure.

Examples: emails, XML, markup languages like HTML, JSON.

Features of semi-structured data (5 points)

1. It **does not conform to the data models** associated with relational databases or any other form of data tables.
2. It uses tags to segregate semantic elements.
3. Tags are also used to **enforce hierarchies** of records and fields within data.
4. There is no separation between the data and the schema. The amount of structure used depends on the purpose at hand.
5. Entities of the **same class** grouped together need **not** have the same set of attributes. Even if they do, the **order** of attributes may differ — and it doesn't matter.

Characteristics of semi-structured data

🔀 Inconsistent structure

Records may look different from each other.

🏷 Self-describing

Uses **label/value pairs** (e.g., `"name": "Asha"`).

🧩 Schema blended with data

Schema information is often mixed into the data values.

❓ Unknown attributes

Data objects may have different attributes **not known beforehand**.

Sources of semi-structured data

The front runners are XML and JSON (plus other markup languages like HTML).

1. **XML — eXtensible Markup Language**: hugely popularised by web services built on the **SOAP** (Simple Object Access Protocol) principles.
2. **JSON — JavaScript Object Notation**: used to **transmit data between a server and a web application**. Popularised by web services built on **REST** (Representational State Transfer). **MongoDB** and **Couchbase** (originally Membase) — open-source, distributed, NoSQL, document-oriented databases — store data **natively in JSON**.
3. **Other markup languages** — e.g., HTML.

⚠️ **DON'T MIX UP**

XML ↔ SOAP
JSON ↔ REST
MongoDB / CouchDB / Couchbase ↔ JSON
These pairs appear in "match the following".

Example — HTML (a markup language)

```
<HTML>
 <HEAD>
  <TITLE>Place your title here</TITLE>
 </HEAD>
 <BODY BGCOLOR="FFFFFF">
  <CENTER><IMG SRC="clouds.jpg"
     ALIGN="BOTTOM"></CENTER>
  <a href="http://bigdatauniversity.com">Link
Name</a>
  <H1>this is a Header</H1>
  <H2>this is a sub Header</H2>
  <P>a new paragraph
 </BODY>
</HTML>
```

Example — a JSON document

```
{
  _id: 9,
  BookTitle: "Fundamentals of
              Business Analytics",
  AuthorName: "Seema Acharya",
  Publisher: "Wiley India",
  YearofPublication: "2011"
}
```

The **yellow** keys are the "tags" (labels). Each value carries its own label → **self-describing**.

💡 **WHY IS AN EMAIL "SEMI-STRUCTURED"?**

Its **header** has fixed labelled fields (To, From, Subject, Date) → some structure. But its **body** is free text → unstructured. So an email as a whole is semi-structured, while the *body of an email* is listed as unstructured.

1.5 Unstructured data

DEFINITION

Unstructured data is data which does not conform to a data model or is not in a form which can be used easily by a computer program. About **80–90%** of an organisation's data is in this format.

Examples: memos, chat rooms, PowerPoint presentations, images, videos, letters, research papers, white papers, body of an email.

Characteristics

- **Un-organised** data
- **No predefined data model** (non-relational)
- 80% of enterprise data is unstructured (Source: Gartner)
- Can **never be stored** in databases' tables, rows and columns
- **Difficult to search**
- Storing **AADHAR images** of users is unstructured data
- Examples: social media data, images, videos, CCTV data, etc.

Sources of unstructured data

Web pages | Images | Free-form text | Audios | Videos | Body of email | Text messages | Chats | Social media data | Word documents

Forms of unstructured data: machine-generated vs human-generated

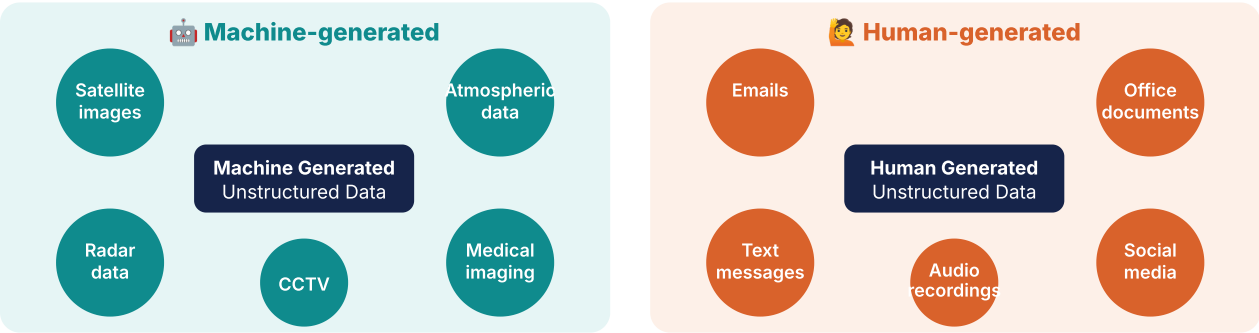


Figure 1.3 — Forms/sources of unstructured data

Few examples of disparate unstructured data (Textbook Table 1.4)

Source	Example
Twitter message	Feeling miffed ☹. Victim of twishing.
Facebook post	LOL. C ya. BFN
Log files	`127.0.0.1 - frank [10/Oct/2000:13:55:36 -0700] "GET /apache_pb.gif HTTP/1.0" 200 2326 "http://www.example.com/start.html" "Mozilla/4.08 [en] (Win98; I; Nav)"`
Email	Hey Joan, possible to send across the first cut on the Hadoop chapter by Friday EOD or maybe we can meet up over a cup of coffee. Best regards, Tom

Issues with terminology

Calling data "unstructured" isn't always perfect. Three issues:

1. **Structure can be implied** despite not being formally defined.
2. Data with some structure may **still be labelled unstructured** if the structure doesn't help with the processing task at hand.
3. Data may have some structure or may even be **highly structured in ways that are unanticipated** or unannounced.

1.6 Dealing with unstructured data

How do we find patterns in or interpret unstructured data? The textbook lists these **techniques**:



Figure 1.4 — Techniques to find patterns in / interpret unstructured data

🧠 **MEMORY TRICK — 7 TECHNIQUES**

"Dark Tea Never Needs More Pure Upgrades" → **D**ata mining · **T**ext analytics · **N**LP · **N**oisy text analytics · **M**anual tagging with metadata · **P**OS tagging · **U**IMA.

⛏ 1. Data Mining

Finding **useful patterns** or information from a large amount of data. Also called knowledge discovery in databases (KDD), because we discover useful knowledge from stored data. Uses methods at the intersection of **statistics, AI, machine learning and databases**.

e.g., a supermarket analyses thousands of bills to find: what products people buy together, which products are popular, how one factor affects another.

Popular algorithms:

- **Association rule mining** — relationships between items: customers who buy *bread* also buy *butter* (market basket analysis).
- **Regression analysis** — predicts one variable from another: predicting marks from study hours.
- **Collaborative filtering** — predicts what a person will like based on what **similar people** like (Netflix/Amazon recommendations).

📝 2. Text Analytics / Text Mining

Finding useful information and patterns **from text** — emails, blogs, social media, messages, reviews.

e.g., analysing 10,000 product reviews → "Most customers are happy, but many complain about battery life."

Text mining includes:

- **Text categorisation** → putting text into categories
- **Text clustering** → grouping similar text
- **Sentiment analysis** → positive / negative / neutral
- **Concept/entity extraction** → finding names, places, organisations

💬 3. Natural Language Processing (NLP)

Teaching computers to understand human (natural) language.

Examples: Google Translate, chatbots, voice assistants, spam detection, speech-to-text. You type "I am feeling happy today." → NLP helps the computer understand its meaning.

🤪 4. Noisy Text Analytics

Noisy text = text with mistakes, abbreviations, slang or unnecessary words — e.g., *"hey!!! u coming tmrw??"*, *"I ammmmm sooooo happyyyy"*. Noisy text analytics extracts useful info from such messy text (chats, SMS, blogs, call-centre notes).

It deals with: spelling mistakes, abbreviations (u = you), slang, missing punctuation, missing letters, unnecessary words like "wru".

🏷 5. Manual Tagging with Metadata

Metadata = information about data. Manual tagging = a **person** manually adds labels/tags so computers can understand and organise the information.

A news article tagged: Sports, Cricket, India, 2026 → easier to organise & search.

🔤 6. Part-of-Speech (POS) Tagging

Identifying the **grammatical role** of each word in a sentence.

"Rahul plays cricket." → Rahul = Noun, plays = Verb, cricket = Noun.
"The beautiful flower grows." → The = Determiner, beautiful = Adjective, flower = Noun, grows = Verb.

🧱 7. UIMA — Unstructured Information Management Architecture

An open-source platform originally developed by IBM. Its purpose is to **process unstructured data (such as text) and extract useful information** from it — used for real-time **content analytics**.

"Apple launched a new product in California." → Apple = Organisation, California = Location, new product = Concept/entity.

🎯 **EXAM TIP**

"Explain the techniques to find patterns in or interpret unstructured data" is a **sample question on your slides**. Draw Figure 1.4, then write 2–3 lines + one example for each technique.

1.7 Comparison: Structured vs Semi-structured vs Unstructured

Property	Structured data	Semi-structured data	Unstructured data
Technology	Based on **relational database tables**	Based on **XML / RDF** (Resource Description Framework)	Based on **character and binary** data
Transaction management	**Matured** transaction and various concurrency techniques	Transaction is adapted from DBMS, **not matured**	**No** transaction management and no concurrency
Version management	Versioning over tuples, rows, tables	Versioning over tuples or graph is possible	Versioned **as a whole**
Flexibility	**Schema dependent**, less flexible	More flexible than structured, less flexible than unstructured	**Most flexible** — absence of schema
Scalability	**Very difficult** to scale DB schema	Scaling is simpler than structured	**Most scalable**
Robustness	**Very robust**	New technology, not very widespread	—
Query performance	Structured query allows **complex joining**	Queries over anonymous nodes are possible	**Only textual queries** are possible
Examples	RDBMS tables, Excel, MS Access	XML, JSON, HTML, email	Images, video, audio, chats, social posts

🧠 **MEMORY TRICK — 7 PROPERTIES**

"The Tiny Vampire Flew Swiftly, Rarely Querying" → **T**echnology · **T**ransaction mgmt · **V**ersion mgmt · **F**lexibility · **S**calability · **R**obustness · **Q**uery performance.
Pattern to remember: going **Structured → Semi → Unstructured**, **flexibility & scalability go UP** while **robustness, transactions & query power go DOWN**.

1.8 Sample questions — solved

A. Place me in the basket

Words: Email, MS Access, Images, Database, Chat conversations, Relations/Tables, Facebook, Videos, MS Excel, XML

Structured	Semi-structured	Unstructured
MS Access Database Relations / Tables MS Excel	XML Email (header has fields; body is free text)	Images Videos Chat conversations Facebook

B1. Match the following

Column A	Column B (answer)
NLP	Comprehend human or natural language input
Text analytics	Text mining
UIMA	Content analytics
Noisy unstructured data	Text messages
Data mining	Uses methods at the intersection of statistics, AI, ML & DBs
Noisy unstructured data	Chats
IBM	UIMA

B2. Match the following

Column A	Column B (answer)
JSON	REST
MongoDB	JSON
XML	SOAP
Flexible structure	XML
JSON	CouchDB

C. Short answers

Q1. Why is an email placed in the "unstructured" category?

The **body of an email** is free-form text with no data model — it can contain anything (sentences, slang, links, attachments, images). It can't be stored in rows and columns or queried easily, so it's unstructured. (Its *header* — To, From, Date, Subject — does have structure, which is why emails are sometimes called semi-structured.)

Q2. A book is issued from the library. What details can be placed in an RDBMS table?

Book(BookID *PK*, Title, Author, Publisher, ISBN, Edition, YearOfPublication, Category) · **Member**(MemberID *PK*, Name, Department, Phone) · **Issue**(IssueID *PK*, BookID *FK*, MemberID *FK*, IssueDate, DueDate, ReturnDate, Fine). Constraints: BookID UNIQUE/NOT NULL, DueDate > IssueDate, Fine ≥ 0.

Q3. Define digital data and explain the sources of data.

Use the definition from §1.1 + internal vs external sources (§1.1 and §2.5) + homogeneous vs heterogeneous sources. Mention the three forms: structured, semi-structured, unstructured.

Q4–Q8. Classification of digital data · Structured data with example · Note on semi-structured & unstructured · Working of structured data · Techniques for unstructured data

See §1.2 (Fig 1.1), §1.3 (Employee table + Fig 1.2 + ease of working + ACID), §1.4 & §1.5, and §1.6 (Fig 1.4). Always add the diagram!

✅ **Before you move on — can you…**

- ☐ Define data, and draw the "Types of data" tree (quantitative/qualitative)?
- ☐ Name the 3 types of digital data with 2 examples each, and say what % is unstructured?
- ☐ Explain cardinality vs degree, and the 5 eases of structured data (ISIST) incl. ACID?
- ☐ List the 5 features + 4 characteristics of semi-structured data; pair XML–SOAP, JSON–REST?
- ☐ Draw machine vs human-generated unstructured data and list the 7 techniques to deal with it?
- ☐ Reproduce the 7-row comparison table?

CHAPTER 2

Introduction to Big Data

What you'll learn

1. Characteristics of data (3 Cs)
2. Evolution of Big Data
3. Definition of Big Data (Gartner)
4. Challenges with Big Data
5. The 3 Vs: Volume, Velocity, Variety
6. Other Vs: Veracity, Volatility, Variability
7. Why Big Data?
8. Traditional BI vs Big Data

⏱ ≈ 1.5 hours | Slides: Lecture 2 (pp. 48–70) | Textbook Ch. 2

2.1 Characteristics of data

🗣 **IN SIMPLE WORDS**

Before analysing any data, ask 3 questions: **What is it made of?** (Composition) · **Is it in good shape?** (Condition) · **Where and why did it come from?** (Context).

Data has **three characteristics**:

1. **Composition** — deals with the structure of data: the **sources** of data, the **granularity**, the **types**, and the **nature** of data — whether it is **static or real-time streaming**.
2. **Condition** — deals with the state of the data: "Can one use this data **as is** for analysis?" or "Does it require **cleansing** for further enhancement and enrichment?"
3. **Context** — deals with "Where has this data been generated?", "Why was this data generated?", and so on.

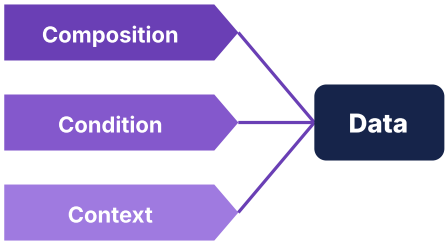


Figure 2.1 — Characteristics of data

🧠 **MEMORY TRICK**

The **3 Cs** of data: **C**omposition (what), **C**ondition (how good), **C**ontext (where & why).

2.2 Evolution of Big Data

Big Data did not appear overnight. Data grew in three eras:

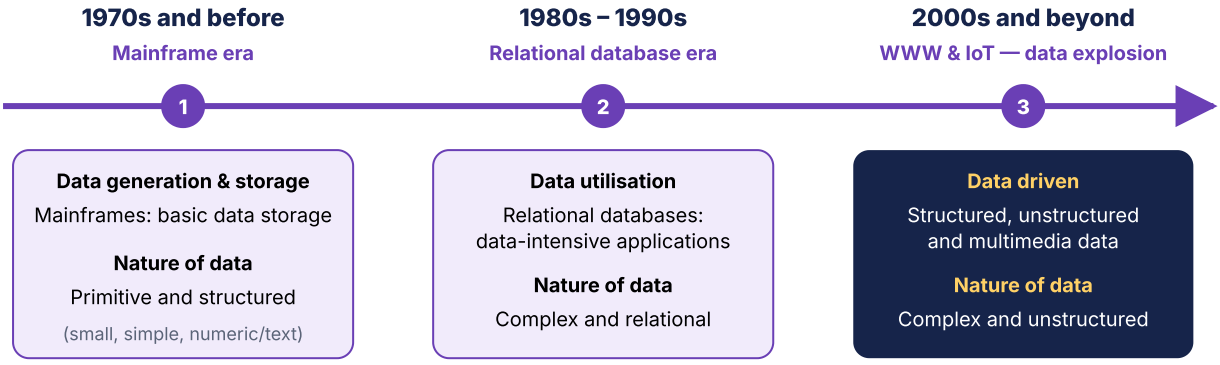


Figure 2.2 — Evolution of Big Data (data generation & storage → data utilisation → data driven)

🔑 **KEY POINT**

Data moved from primitive & structured (mainframes) → complex & relational (RDBMS) → complex & unstructured (WWW, IoT, social media). The **web and IoT** caused the "explosion of data" that we now call Big Data.

2.3 Definition of Big Data

People describe Big Data in different informal ways (from your slides):

1. "**Anything beyond the human and technical infrastructure** needed to support storage, processing, and analysis."
2. "**Today's BIG may be tomorrow's NORMAL.**" (what is "big" keeps changing)
3. "**Terabytes or petabytes or zettabytes** of data."
4. "I think about the **3 Vs**." (Volume, Velocity, Variety)

GARTNER DEFINITION — LEARN WORD-FOR-WORD

"Big Data is high-volume, high-velocity and high-variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making."

Source: Gartner IT Glossary

The definition has **three parts** — learn them as a cycle:

Part 1 · What it is
High-volume, high-velocity, high-variety information assets

→

Part 2 · What it needs
Cost-effective, innovative forms of information processing

→

Part 3 · What it gives
Enhanced insight & decision making

Data → **Information** → **Actionable intelligence** → **Better decisions** → **Enhanced business value**

🎯 EXAM TIP

"Define Big Data" → write the **Gartner definition**, draw the 3-part cycle, then add one line on each of the **3 Vs**. That's a complete 5-mark answer.

2.4 Challenges with Big Data

1. **Huge growth of data** — data is growing **exponentially**, so it is difficult to decide which data is **useful** and which is **noise**. We must select relevant data for analysis.
2. **Cloud computing & virtualisation** — cloud helps manage Big Data infrastructure efficiently: **cost-effective, flexible**, easy **upgrading or downgrading** of resources. (Challenge: deciding whether/how to host data on the cloud.)
3. **Data retention** — deciding **how long** to store Big Data. Some data is useful for long-term decisions; some becomes irrelevant quickly.
4. **Lack of skilled professionals** — shortage of experts in data science and Big Data technologies → hard to implement Big Data solutions.
5. **Data capture, storage, processing & security** — challenges in capturing, storing, preparing, searching, analysing, transferring, securing and visualising data. Traditional databases may not handle data that is very **large, fast and varied**.
6. **Data visualisation** — representing Big Data with graphs, charts, dashboards helps people see patterns. There is a **shortage of skilled visualisation professionals** too.

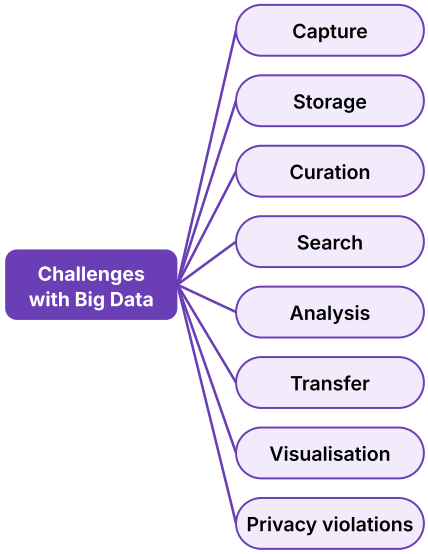


Figure 2.3 — Challenges with Big Data

🧠 MEMORY TRICK — 6 CHALLENGES

"Giant Clouds Rarely Lack Some Views" → **G**rowth of data · **C**loud & virtualisation · **R**etention · **L**ack of skilled people · **S**torage/capture/processing/security · **V**isualisation.

⚠️ DON'T CONFUSE

"**Challenges with Big Data**" (this list, Ch. 2) ≠ "**Top challenges facing Big Data**" (Scale, Security, Schema, Availability, Consistency, Partition tolerance, Data quality — Ch. 3, §3.4). Read the question carefully!

2.5 What is Big Data? — the 3 Vs

DEFINITION

Big data is data that is big in volume, velocity, and variety.

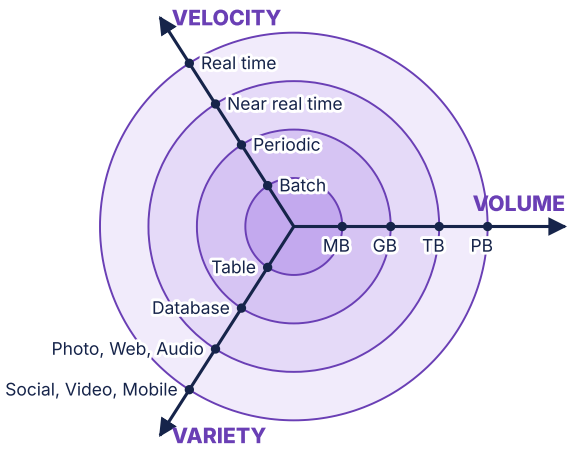


Figure 2.4 — Data: big in volume, velocity and variety

📦 Volume — how MUCH data

Huge amounts — from GB/TB to PB, EB, ZB. Comes from internal & external sources.

⚡ Velocity — how FAST data arrives

Batch → Periodic → Near real time → Real-time processing.

🎨 Variety — how many KINDS of data

Structured + semi-structured + unstructured, from many sources.

🧠 MEMORY TRICK

Imagine a **river**: **Volume** = how much water, **Velocity** = how fast it flows, **Variety** = how many different things are floating in it.

Volume — the growth of data

Unit	Abbr.	Size (binary, textbook)	≈ Decimal ("mountain of data")	Example use case
Bit	b	0 or 1	—	Binary value
Byte	B	8 bits	—	A single character
Kilobyte	KB	1024 bytes	1,000 bytes (10^3)	Small text files
Megabyte	MB	1024^2 bytes	10^6 bytes	Images, small MP3 files
Gigabyte	GB	1024^3 bytes	10^9 bytes	Videos, large software applications
Terabyte	TB	1024^4 bytes	10^{12} bytes	Hard drives, storage in modern laptops
Petabyte	PB	1024^5 bytes	10^{15} bytes	Daily data processed by large internet companies
Exabyte	EB	1024^6 bytes	10^{18} bytes	Monthly global internet traffic

Unit	Abbr.	Size (binary, textbook)	≈ Decimal ("mountain of data")	Example use case
Zettabyte	ZB	1024^7 bytes	10^{21} bytes	Estimated digital universe size by 2025
Yottabyte	YB	1024^8 bytes	10^{24} bytes	Hypothetical large-scale data storage

🧠 **MEMORY TRICK — ORDER OF UNITS**

"Kind Men Give Tea, Please Eat Zesty Yogurt" → KB · MB · GB · TB · PB · EB · ZB · YB. Each step is **×1024** (≈ ×1000). Power of 1024 = position in the list (KB = 1024^1 … YB = 1024^8).

Where does this volume come from? — Sources of Big Data

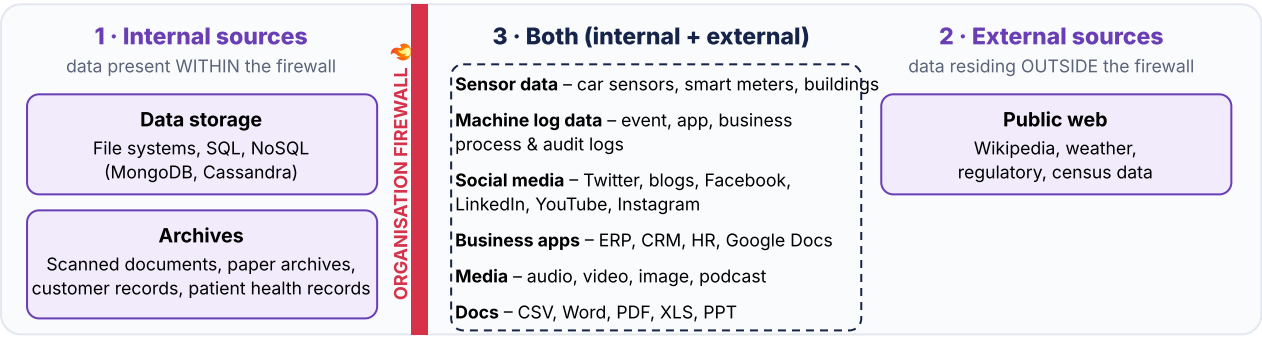


Figure 2.5 — Sources of Big Data: internal, external and both

Velocity — the speed of data

Batch
collected, then processed together later
e.g., monthly payroll

→

Periodic
processed at fixed intervals
e.g., hourly/daily reports

→

Near real time
processed within seconds/minutes
e.g., traffic updates on maps

→

Real-time
processed instantly as it arrives
e.g., card-fraud check at swipe

Variety — the kinds of data

Variety deals with a **wide range of data types and sources**:

1. **Structured data**: from traditional transaction processing systems and RDBMS, etc.
2. **Semi-structured data**: e.g., HyperText Markup Language (HTML), eXtensible Markup Language (XML).
3. **Unstructured data**: e.g., unstructured text documents, audios, videos, emails, photos, PDFs, social media, etc.

2.6 Other characteristics of data

These are not definitional traits of Big Data (the definition only needs the 3 Vs), but are still important:

✅ Veracity & Validity

Veracity = trustworthiness of data (is there noise, bias or abnormality?).
Validity = accuracy and correctness of data for the given problem.

e.g., fake reviews reduce veracity.

⏳ Volatility

How long the data remains valid and useful. Helps decide **storage duration** — some data stays useful for long, some becomes obsolete quickly.

e.g., today's stock price is stale tomorrow; birth date never expires.

📉 Variability

Data flow is inconsistent and can have **sudden peaks**. Systems must handle such irregular, fluctuating data rates.

e.g., traffic spike on a shopping site during a festival sale.

🔑 **KEY POINT**

Core Big Data = **3 Vs** (Volume, Velocity, Variety). The "extra" characteristics = **Veracity & Validity, Volatility, Variability** — together often called the **6 Vs**.

⚠️ **COMMON MISTAKE**

Velocity (speed of arrival) ≠ **Variability** (inconsistency of the flow) ≠ **Volatility** (how long data stays useful). Learn one example for each.

2.7 Why Big Data?

More data → **More accurate analysis** → **More confidence in decision making** → **Greater operational efficiencies**
cost reduction, time reduction, new product development, optimised offerings, etc.

The importance of Big Data lies in how you utilise the data which you own. Data can be fetched from any source and analysed to enable:

💰 Cost reductions

⏱️ Time reductions

🚀 New product development & optimised offerings

🧠 Smart decision making

2.8 Traditional Business Intelligence (BI) vs Big Data

🗣️ **IN SIMPLE WORDS**

Business Intelligence (BI) = the old way: a company keeps its own neat (structured) data on one big central server and makes reports about the past. **Big Data** = the new way: data of all kinds, from inside and outside, spread over many cheap machines, analysed in real time too.

Three textbook differences (write these in a long answer):

1. In traditional BI, all the enterprise's data is housed in a central server, whereas in Big Data it resides in a distributed file system. The distributed file system **scales horizontally** (scaling in or out), compared to a typical database server that **scales vertically**.
2. In traditional BI, data is generally analysed in an offline mode; in Big Data, it is analysed in both real time and offline mode.
3. Traditional BI is about **structured data**, and here data is taken to the processing functions; Big Data is about **variety**, and here the processing functions are taken to the data.

Point	Traditional BI	Big Data
Storage	Central server	Distributed file system
Scaling	Vertical (bigger machine) — "scale up"	Horizontal (add more machines) — "scale out"
Analysis mode	Mostly offline	Real-time + offline
Data type	Structured	Structured + unstructured (variety)
Processing style	Data moved to processing functions	Processing functions moved to the data
Volume	Limited (GBs)	Very large (TBs/PBs)
Data sources	Mostly internal	Internal + external
Schema	Schema-on-write	Schema-on-read
Cost	High (expensive servers)	Lower (commodity hardware)

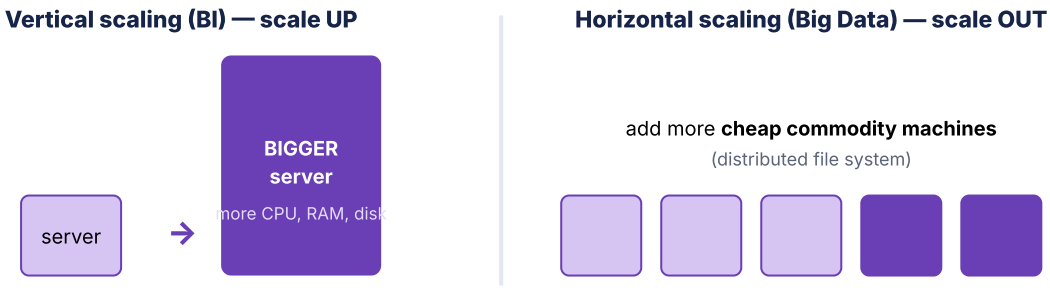


Figure 2.6 — Scale up (traditional BI) vs scale out (Big Data)

💡 **SCHEMA-ON-WRITE VS SCHEMA-ON-READ**

Schema-on-write (BI): you must define the table structure *before* storing data — data that doesn't fit is rejected. **Schema-on-read** (Big Data): store the raw data first in any form; apply a structure only *when you read/analyse* it. This is why Big Data handles variety easily.

✅ **Before you move on — can you...**

☐ Explain the 3 Cs (composition, condition, context) with the diagram?

☐ Draw the evolution timeline (mainframe → RDBMS → WWW/IoT)?

☐ Write the Gartner definition word-for-word and its 3 parts?

☐ List the 6 challenges with Big Data + draw the 8-bubble challenges diagram?

☐ Explain the 3 Vs with the concentric-circle figure, list the units KB → YB, and internal vs external sources?

☐ Explain veracity, validity, volatility, variability with an example each?

☐ Write the BI vs Big Data differences (3 textbook points + table)?

CHAPTER 3

Big Data Analytics

What you'll learn

1. Analytics through examples
2. What BDA is — and isn't
3. Classification: 1.0 / 2.0 / 3.0
4. Top challenges facing Big Data
5. Why BDA is important
6. Data Science & its 3 skill sets
7. Data Scientist responsibilities
8. Terminologies: in-memory, in-database, SMP, MPP, shared nothing, CAP

⏱ ≈ 3 hours (two sessions) | Slides: Lectures 3–4 (pp. 71–140) | Textbook Ch. 3

3.1 Understanding Big Data Analytics through examples

🛒 Scenario 1 — Amazon recommendation system

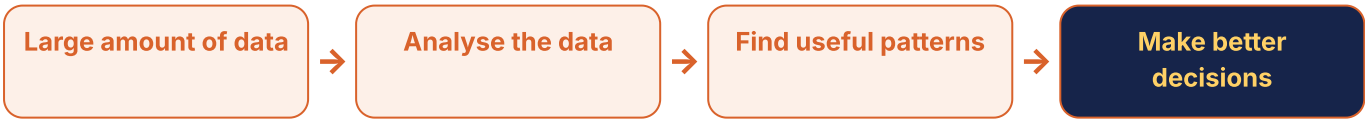
You register on Amazon and buy a book by your favourite author. Soon Amazon recommends: books by the **same author**, books on **similar topics**, products based on your **previous purchases**, and products bought by **customers with similar interests**.

How? Amazon collects and analyses your **past purchases, browsing history, preferences and buying patterns**.

🚚 Scenario 2 — Truck transport company

You own **500 trucks**. A customer asks for an **urgent delivery** and will pay **double**. Which truck should you send?

You must analyse: **current truck location, available load capacity, fuel consumption, traffic on routes, distance to customer, expected delivery time**.



Key idea of Big Data Analytics

Where do we begin with Big Data Analytics?

Big Data Analytics starts with **raw data**. Raw data by itself may not be very useful — we process it step by step until it is useful for decisions:

1. **Collecting → DATA**: collect raw data from different sources.
2. **Organising → INFORMATION**: classify and organise data so it becomes meaningful.
3. **Summarising & Analysing → KNOWLEDGE**: find patterns, trends and relationships.
4. **Synthesising → ACTIONABLE INSIGHTS**: decide what useful action can be taken.
5. **DECISION MAKING**: use the insights to make a better and faster decision.

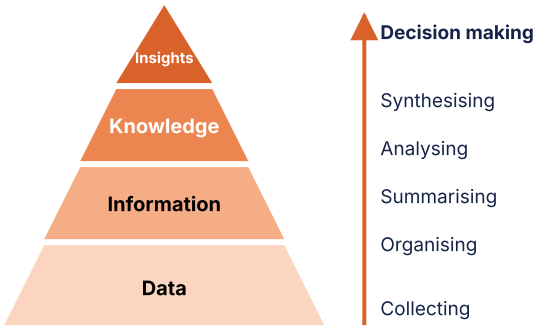


Figure 3.1 — Transformation of data to yield actionable insights

Where does the data come from?

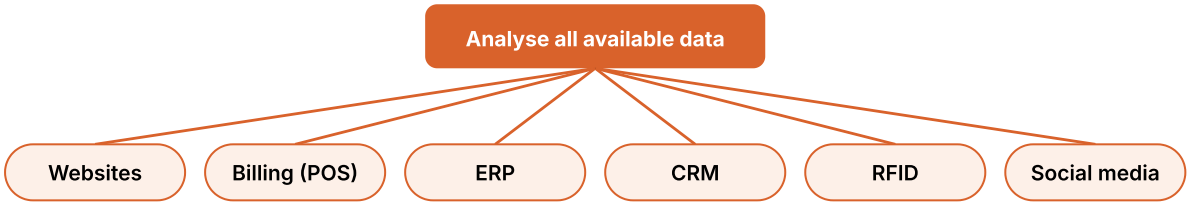


Figure 3.2 — Types of data available for analysis

POS = Point of Sale · ERP = Enterprise Resource Planning · CRM = Customer Relationship Management · RFID = Radio Frequency Identification

3.2 What is Big Data Analytics?

IN SIMPLE WORDS

Big Data Analytics (BDA) = using tools to dig through **huge, varied, fast** data to find **patterns** that help a business decide **better and faster** than competitors.

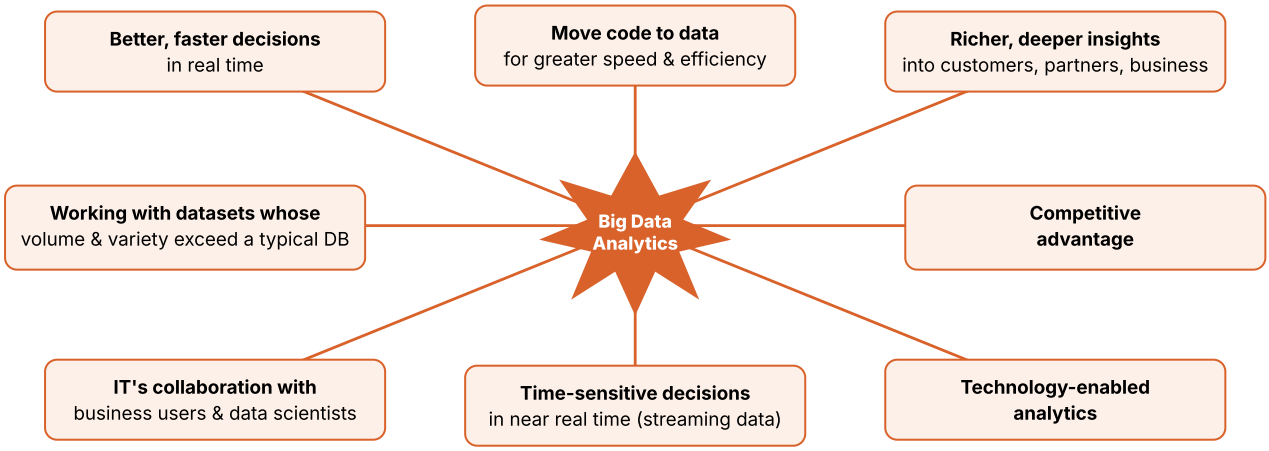


Figure 3.3 — What is Big Data Analytics?

Your slides explain **6 points**:

1. Technology-enabled analytics

Many analytics and visualisation tools are available to process huge data — from **IBM, Tableau, SAS, R Analytics, statistical tools, WPS** (World Programming Systems).

2. Getting deeper, meaningful insights

Having data is not enough — we need **insight**. Helps a company understand: who its customers are, what they like and buy, what else can be sold to them (cross-sell), how to use vendors and suppliers effectively.

3. Gives a competitive edge

Quicker and better decisions than competitors. Company A analyses customer behaviour and offers the right products early; Company B doesn't → A wins customers.
Better analysis → Faster decisions → Competitive advantage.

4. A tight handshake between three communities

IT team — provides computers, databases, networks, infrastructure.
Data scientists — analyse data and discover patterns.
Business users — use the results to make business decisions.

5. Working with very large & varied datasets

Volume (huge quantity) and **variety** (customer records + images + videos + emails + social posts + transactions) may **exceed the capability of traditional systems**.

6. Moving code to data

Program = a few KB/MB; data = TB/PB. It's easier to move the **small program to where the data is stored** than to move huge data to the program. Processing happens **close to the data**.



Figure 3.4 — "Moving code to data"

🧠 **MEMORY TRICK — 6 POINTS OF "WHAT IS BDA"**

"The Deep Competitive Handshake With Movement" → **T**echnology-enabled · **D**eeper insights · **C**ompetitive edge · **H**andshake of 3 communities · **W**orking with large varied data · **M**oving code to data.

What Big Data Analytics isn't

❌ Only about **volume**

❌ Just about **technology**

❌ Meant to **replace RDBMS**

❌ Meant to **replace the data warehouse**

❌ Only used by huge online companies like **Google or Amazon**

❌ **"One-size-fits-all"** traditional RDBMS built on shared disk and memory

Why this sudden hype around Big Data Analytics?

📈 1. Huge growth of data

Data is exploding from web, mobiles, sensors, social media.

💾 2. Storage has become cheaper

Storing TBs/PBs is now affordable (commodity hardware, cloud).

🛠 3. Availability of analytics tools

Open-source and commercial tools (Hadoop, R, Tableau...) are easily available.

3.3 Classification of analytics

There are basically **two schools of thought**:

First school of thought — 4 types

1. **Basic analytics**: slicing and dicing of data for basic business insights — reporting on **historical data**, basic visualisation, etc.
2. **Operationalised analytics**: analytics that gets **woven into the enterprise's business processes**.
3. **Advanced analytics**: **forecasting the future** using predictive and prescriptive modelling.
4. **Monetised analytics**: analytics used to derive **direct business revenue**.

Second school of thought — 3 generations

Analytics 1.0 → Analytics 2.0 → Analytics 3.0

Each generation **adds** a new type of analytics on top of the previous one:

1.0 = Descriptive
2.0 = Descriptive + Predictive
3.0 = Descriptive + Predictive + Prescriptive

🧠 **MEMORY TRICK — FIRST SCHOOL**

"BOAM" — **B**asic, **O**perationalised, **A**dvanced, **M**onetised (from *looking back → making money*).

Analytics 1.0, 2.0 and 3.0

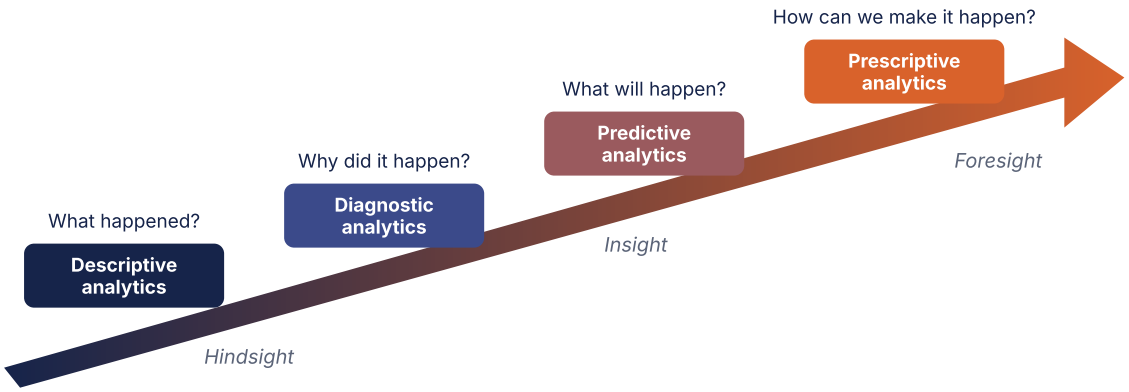


Figure 3.5 — Analytics 1.0, 2.0 and 3.0: from hindsight to foresight

	Analytics 1.0	Analytics 2.0	Analytics 3.0
Era	Mid 1950s to 2009	2005 to 2012	2012 to present
Main idea	Understand the past	Predict the future	Predict & recommend
Type of analytics	**Descriptive** statistics (report on events, occurrences of the past)	**Descriptive + predictive** statistics (use past data to make predictions)	**Descriptive + predictive + prescriptive** (predict the future and recommend how to leverage the situation to one's advantage)
Key questions	What happened? Why did it happen?	What will happen? Why will it happen?	What will happen? When will it happen? Why will it happen? What action should be taken to take advantage of it?
Data	From legacy systems, ERP, CRM, 3rd-party apps. **Small & structured**, stored in enterprise data warehouses / data marts	**Big data** taken seriously — mainly **unstructured**, arriving fast; stored & processed rapidly on massive parallel servers running **Hadoop**	**A blend** of big data and data from legacy systems, ERP, CRM & 3rd-party apps; big data + traditional analytics for speed and impact
Source	Internally sourced	Often externally sourced	Both internally & externally sourced
Technology	Relational databases	Database appliances, Hadoop clusters, SQL-to-Hadoop environments	In-memory analytics, in-database processing, agile analytical methods, machine-learning techniques
Simple example	"Sales fell last month."	"Sales may rise next month."	"Increase stock next month."

🎯 **EXAM TIP**

"Explain the classification of analytics" → write **both schools**. For the second school, draw the **staircase (Fig. 3.5)** and the table with at least 5 rows: era, type, key questions, data, technology.

🧠 **MEMORY TRICK — THE 4 ANALYTICS**

"Do Doctors Prescribe Properly?" in order: **D**escriptive (what happened) → **D**iagnostic (why) → **P**redictive (what will) → **P**rescriptive (what should we do). **Hindsight → Insight → Foresight.**

3.4 Top challenges facing Big Data

📏 1. Scale

How to **store and manage** the huge and continuously growing amount of data.

🔐 2. Security

How to **protect** Big Data from unauthorised access and attacks.

🧬 3. Schema

Rigid schemas have no place in Big Data — we need **dynamic schemas**.

🕐 4. Continuous availability

Systems must give **24/7 support** because downtime is a problem.

🎯 5. Consistency

Make sure users see **correct and updated** data.

🧱 6. Partition tolerant

How to build systems that keep working despite **hardware and software failures**.

✨ 7. Data quality

How to maintain data **accuracy, completeness, timeliness**, and proper **metadata**.

🧠 **MEMORY TRICK — 7 TOP CHALLENGES**

"Six Students Sat Calmly, Copying Past Quizzes" → **S**cale · **S**ecurity · **S**chema · **C**ontinuous availability · **C**onsistency · **P**artition tolerance · **Q**uality of data.
Notice: Consistency + Availability + Partition tolerance = **CAP** (see §3.9).

3.5 Why is Big Data Analytics important?

Big Data Analytics is important because it converts huge amounts of data into useful information, helping organisations **make better decisions, predict future events, and solve complex problems**. The textbook explains it through **4 approaches** to analysis:

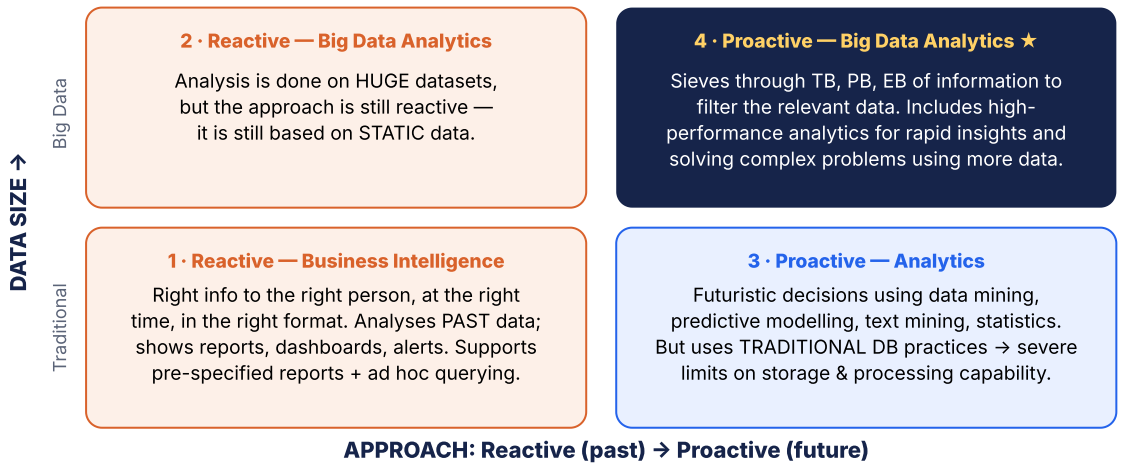


Figure 3.6 — Four approaches to analysis (the goal is box 4: proactive Big Data Analytics)

🔑 KEY POINT

Reactive = looks at the **past** ("what happened?"). **Proactive** = looks at the **future** ("what will happen?"). Only Proactive Big Data Analytics combines **future-looking analysis** with **huge data** — that's why BDA is important.

3.6 Data Science

DEFINITION

Data science is the science of extracting knowledge from data. In simple words, it **finds useful and hidden patterns in large amounts of data**. It uses ideas from **mathematics, statistics and information technology**, and also **machine learning, data engineering, probability models, statistical learning and pattern recognition**.

📌 SIMPLE EXAMPLE — STUDENT EXAM MARKS

DATA	→	PATTERN	→	KNOWLEDGE	→	DECISION
Raw marks of students A–E in Maths, Science, English		Science marks are highest on average; English is lowest		Students do well in Science but need more support in English		Plan extra English practice to improve overall performance

Data Science turns data into knowledge and helps us make better decisions.

Where is Data Science used?

🌦 Weather predictions
Use past & current weather data to predict future weather.

🛢 Oil drilling
Study geological & sensor data to support drilling operations.

🌍 Seismic activities
Analyse earth-movement data to understand earthquakes.

💳 Financial frauds
Detect unusual transactions and suspicious behaviour.

👍 Social media analytics
Study posts, likes and trends to understand users.

🛒 Retail & manufacturing
Customer churn, market basket analysis, filtering & regression.

Data Science is multi-disciplinary

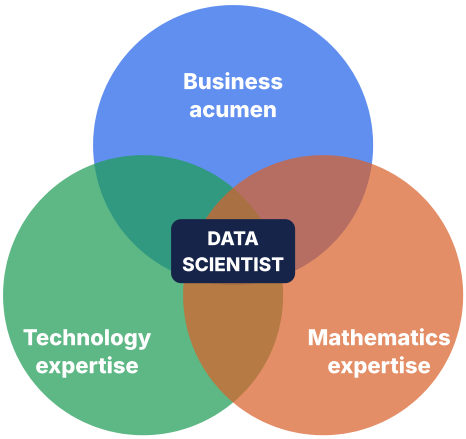


Figure 3.7 — The data scientist sits where all three skill sets overlap

A data scientist needs **all three** skill groups:

- Knowing **only programming** is not enough.
- Knowing **only mathematics** is not enough.
- **Business understanding** helps convert analysis into useful action.

> 🧠 **MEMORY TRICK**
>
> A data scientist is a **"BTM"** person — **B**usiness, **T**echnology, **M**aths. Like a three-legged stool: remove one leg and it falls.

Skill 1 — Business acumen

Business acumen means understanding the real problem and the needs of the organisation. "It's not just about data, it's about real-world impact."

1 · Understanding of domain

Know the field: banking, healthcare, education, retail, etc. Each industry has different problems and needs.

2 · Business strategy

Understand what the organisation wants: increase profit, improve service, reach more customers, reduce cost.

3 · Problem solving

Convert a **business problem** into a solvable **data problem**. "Why are customers leaving?" → analyse customer data to find patterns of churn.

4 · Communication

Explain ideas clearly to others; make complex ideas simple.

5 · Presentation

Show results using simple reports, charts and slides — a good visual helps everyone understand.

6 · Inquisitiveness

Keep asking: Why? What happened? What can we improve? Can we do better?

Skill 2 — Technology expertise

Technical tools help the data scientist **store, process, analyse and present** data.

Area	What it does	Examples
RDBMS	Store and manage data in structured form	MySQL, PostgreSQL, Oracle
NoSQL	Handle large and unstructured data	MongoDB, Cassandra, HBase
Programming	Write code to process and analyse data	Python, Java, C++
Data warehousing	Organise large amounts of data for analysis	Amazon Redshift, Snowflake, BigQuery
Data mining	Discover useful patterns from data	Classification, clustering, association rules
Visualisation	Present data using charts and dashboards	Tableau, Flare, Google Visualization APIs

Skill 3 — Mathematics expertise

A data scientist must understand, interpret and analyse data using mathematical concepts and algorithms. "Maths turns data into meaning."

Area	Role	Examples
Mathematics	Basic language for calculations and modelling	Algebra, calculus, linear algebra
Statistics	Summarise data and make conclusions	Mean, variance, hypothesis testing
Algorithms	Step-by-step methods to solve problems	Sorting, search, optimisation
Machine learning	Learns patterns from data	Regression, classification, clustering
Pattern recognition	Recognises repeated structures or classes	Image recognition, anomaly detection
NLP	Helps computers work with human language	Text classification, sentiment analysis

The Data Science process

Textbook short form: **Collect → Prepare → Analyse → Show → Communicate → Decide**

1 Collect
raw data from multiple sources

2 Process
make raw data usable

3 Integrate & clean
combine sources, prepare a clean dataset

4 Explore & analyse
apply statistics & algorithms

5 Visualise
charts, graphs, infographics

6 Communicate
explain to stakeholders simply

7 Decide
better decisions from insights

Read left → right, top → bottom.

Data + Analysis + Insight = Better Decisions

3.7 Data Scientist & responsibilities

WHO IS A DATA SCIENTIST?

A data scientist is a person who works with data to solve real-world problems. Raw data is often messy; the data scientist **prepares** the data, **studies** it, **finds patterns** and **explains** what they mean. The final goal is not just a graph or a model — it is a useful decision.

A · Raw data
marks, sales, clicks, sensor readings, customer records

→

B · Analysis
clean the data, compare values, find patterns

→

C · Insight
explain what the data tells us and what action to take

Main responsibilities of a data scientist (4)



Figure 3.8 — Data scientist: your new best friend!

🗄 1. Data management

Raw data must be prepared before we can learn from it.

RAW (marks from different classes; values missing/repeated) → **CLEAN** (correct errors, remove duplicates, handle missing values) → **INTEGRATE** (combine marks, attendance, other info) → **READY** (one clean dataset to analyse).

🔍 2. Analytical techniques

Use **models and algorithms** to understand relationships, spot trends, uncover patterns. The technique depends on the **question** and the **type of data**. A model is useful only when it answers the actual question.

Q: Which students may need extra support? → **Analyse** marks + attendance → **Result:** support students earlier.

🎯 3. Business / domain understanding

Numbers alone are not enough — domain knowledge turns a statistical result into a **meaningful solution**.

Cool fact: "Attendance and marks are related." → **Business insight:** "Students with repeated low attendance may need early mentoring." → **Useful action:** mentoring, reminders, extra support.

💬 4. Communication & presentation

A model may produce hundreds of numbers. Convert it into a **chart, key message and simple language** so teachers, managers or customers can act.

Complex analysis → Simple explanation → Stakeholder decision. A good data scientist must be a good presenter and communicator too.

One example — college student performance

1 Collect marks + attendance + assignment data → **2 Prepare** clean and combine the records → **3 Analyse** find relationships and patterns → **4 Understand** connect results to the education context → **5 Communicate** show a simple chart to faculty → **6 Act** plan mentoring or extra classes

Data Science vs Data Scientist

Data Science	Data Scientist
A **FIELD / DISCIPLINE**: methods, processes, maths, statistics, programming, ML	A **PERSON / JOB ROLE**: applies data-science techniques to real problems
e.g., Data Science is used to predict student performance	e.g., A data scientist builds and explains the prediction system

Careers after Data Science

1. **Data Analyst** — clean data, reports, dashboards, trends
2. **Data Scientist** — models, patterns, business problems
3. **ML Engineer** — build & deploy learning systems
4. **Data Engineer** — pipelines that collect & move data
5. **BI / Analytics Professional** — dashboards, business insights
6. **AI / NLP / Vision roles** — language, images, recommenders

🧠 **MEMORY TRICK — 4 RESPONSIBILITIES**

"My Aunt Bakes Cakes" → **M**anage data · **A**nalyse with techniques · **B**usiness/domain context · **C**ommunicate & present.

3.8 Terminologies used in Big Data environments

⚡ In-memory analytics | 🗄 In-database processing | 🧠 SMP & MPP | 🌐 Parallel vs distributed

🔗 Shared nothing architecture | △ CAP theorem

1. In-memory analytics

🗣 IN SIMPLE WORDS

Reading from a hard disk is like walking to the library every time you need a fact. In-memory analytics keeps the books **on your desk (RAM)** — so answers come much faster.

- Data access from **non-volatile storage such as hard disk is a slow process**. The more data is fetched from hard disk/secondary storage, the slower the process gets.
- One way to combat this: **pre-process and store data** (cubes, aggregate tables, query sets, etc.) so the CPU fetches only a small subset of records. But this requires **thinking in advance** about what data will be needed.
- If different or more data is needed, it's back to pre-computing or fetching from secondary storage.
- This problem is solved by in-memory analytics: **all the relevant data is stored in RAM (primary storage)**, eliminating the need to access data from hard disk.



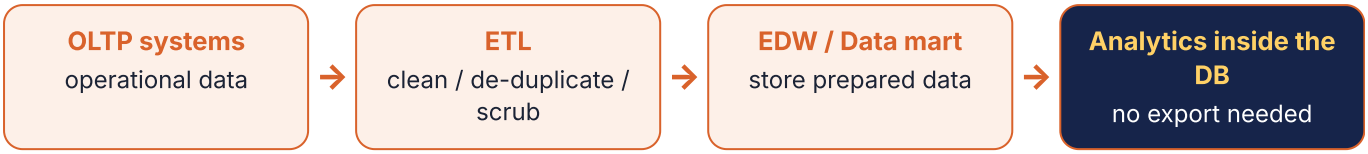
🔑 ADVANTAGES OF IN-MEMORY ANALYTICS

Faster access · **rapid deployment** · **better (quicker) insights** · **minimal IT involvement** (less dependence on repeated disk access).

2. In-database processing

- Also called in-database analytics. It works by **fusing data warehouses with analytical systems**.
- Typically, data from various enterprise **OLTP** (On-Line Transaction Processing) systems — after cleaning up (de-duplication, scrubbing, etc.) through **ETL** — is stored in the **Enterprise Data Warehouse (EDW)** or data marts.
- **Without it**: the huge datasets are then **exported** to analytical programs for complex and extensive computations.

- **With in-database processing**: the database program itself can run the computations, eliminating the need for export and thereby **saving time**. Leading database vendors offer this to large businesses.



ETL = Extract, Transform, Load · EDW = Enterprise Data Warehouse

> 💡 **REMEMBER THE PAIR**
>
> **In-memory** → brings data **closer to the CPU** (RAM instead of disk). **In-database** → brings computation **to where the data lives** (no export). Both reduce **data movement** to save time.

3. Symmetric Multiprocessor System (SMP)

- In SMP there is a single common main memory shared by two or more identical processors.
- The processors have **full access to all I/O devices** and are controlled by a single operating system instance.
- SMPs are tightly coupled multiprocessor systems. Each processor has its own **high-speed memory, called cache memory**, and they are connected using a **system bus**.

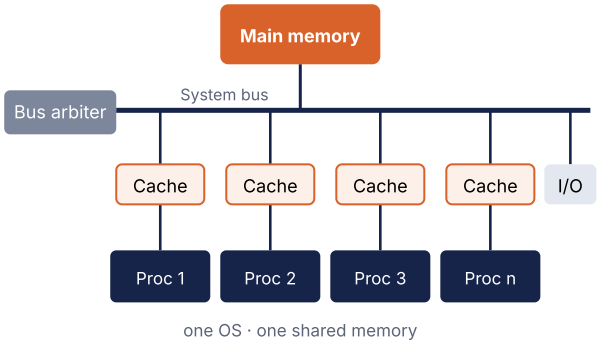


Figure 3.9 — Symmetric Multiprocessor System

4. Massively Parallel Processing (MPP)

- MPP refers to the coordinated processing of programs by a number of processors working in parallel.
- The processors **each have their own operating system and dedicated memory**. They work on **different parts of the same program**.
- MPP processors communicate using some sort of **messaging interface**.
- MPP systems are **more difficult to program**: the application must be divided so that all executing segments can communicate with each other.
- MPP differs from SMP: SMP processors share the same OS and same memory; SMP is also called **tightly-coupled multiprocessing**.

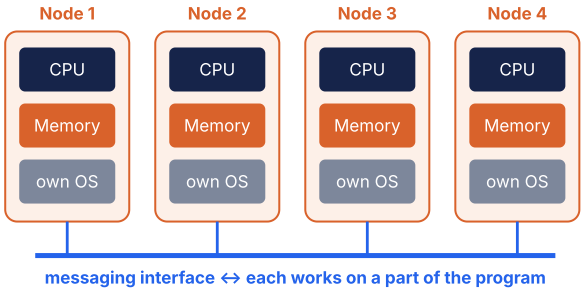


Figure 3.10 — Massively Parallel Processing

Feature	SMP	MPP
Memory	Common / shared main memory	Each node has **dedicated** memory
Coupling	**Tightly** coupled	Looser than SMP; nodes communicate
Operating system	**Single** OS instance	Each processor/node has its **own** OS
Communication	Shared memory / system bus	**Messaging** between nodes
Programming	Easier	Harder — program must be divided into communicating segments

🧠 **MEMORY TRICK**

SMP = "Same Memory, Please" — everyone shares one memory & one OS. **MPP = "My Private Partition"** — each processor has its own memory & OS.

5. Parallel systems vs distributed systems

Parallel systems

- A parallel database system is a tightly coupled system. The processors **co-operate for query processing**.
- The user is **unaware of the parallelism** since he/she has no access to a specific processor of the system.
- Either the processors have access to a **common memory** or make use of **message passing** for communication.

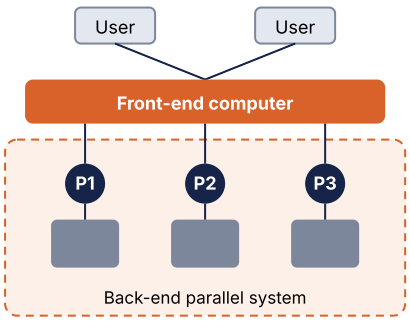


Figure 3.11 — Parallel system

Distributed systems

- Distributed database systems are loosely coupled and composed of **individual machines**.
- Each machine can run its **own application** and serve its **own users**.
- Data is usually **distributed across several machines**, so quite a number of machines may need to be accessed to answer a user query.

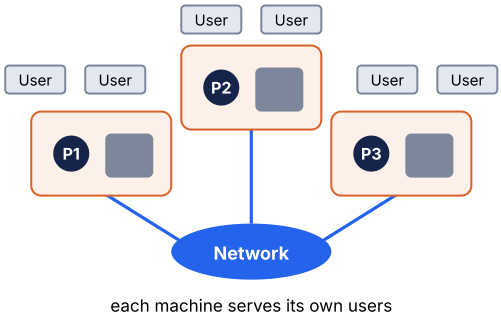


Figure 3.12 — Distributed system

	Parallel system	Distributed system
Coupling	**Tightly** coupled	**Loosely** coupled
Made of	Processors cooperating on the **same problem** (e.g., one large query)	**Separate machines** connected by a network
Memory/communication	Common memory or message passing	Network messages; data spread across machines
User's view	User sees **one system**; unaware of parallelism	Each machine runs its own app and serves its own users; a query may contact several nodes

6. Shared nothing architecture

The three most common types of architecture for **multiprocessor, high-transaction-rate systems**:

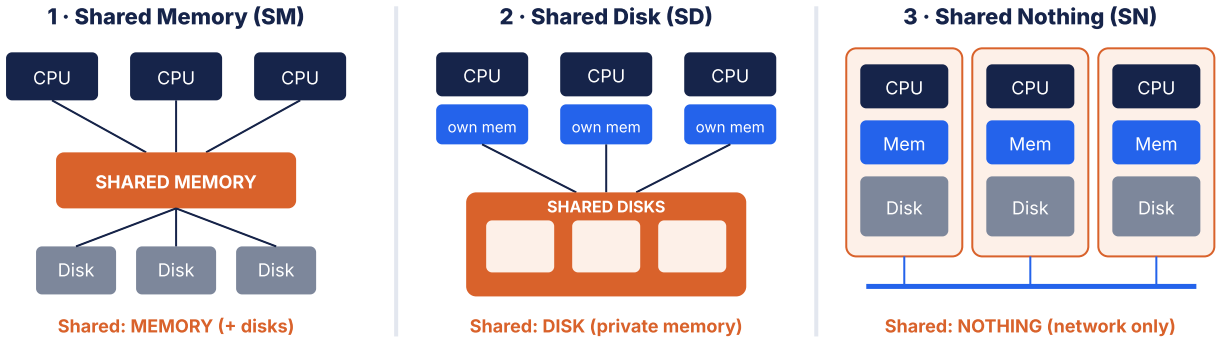


Figure 3.13 — Three multiprocessor architectures

- **Shared Memory (SM)**: a **common central memory** is shared by multiple processors.
- **Shared Disk (SD)**: multiple processors share a **common collection of disks** while having their **own private memory**. Coordination is needed when many nodes access shared data.
- **Shared Nothing (SN)**: neither memory nor disk is shared among processors. Each node has its own processor, memory and storage; nodes communicate through the network.

Advantages of a "Shared Nothing Architecture"

🛡 1. Fault isolation

A fault in a single node is **contained and confined to that node exclusively** and exposed only through messages (or lack of it). It doesn't spread through a common memory/disk.

📈 2. Scalability

If the disk is a shared resource, the controller and disk bandwidth are also shared. **Synchronisation** is needed to keep a consistent shared state, so nodes must **take turns** to access critical data — this **limits how many nodes can be added**. In SN nothing is shared → **add nodes freely**, no central bottleneck.

🎯 EXAM TIP

Hadoop/HDFS (Unit 2!) uses a **shared-nothing** architecture — each data node has its own CPU, memory and disk. Mention this link in answers; it shows understanding.

3.9 CAP theorem

DEFINITION

The CAP theorem is also called Brewer's Theorem. It states that in a **distributed computing environment** (a collection of interconnected nodes that share data), it is **impossible to provide all three** guarantees — Consistency, Availability and Partition tolerance — at the same time. **At best you can have two of the three — one must be sacrificed.**

C · Consistency

Every read fetches the **last write** (latest data).

A · Availability

Reads and writes **always succeed**: each non-failing node returns a response in a reasonable amount of time.

P · Partition tolerance

The system **continues to function** when a **network partition** (communication break between nodes) occurs.

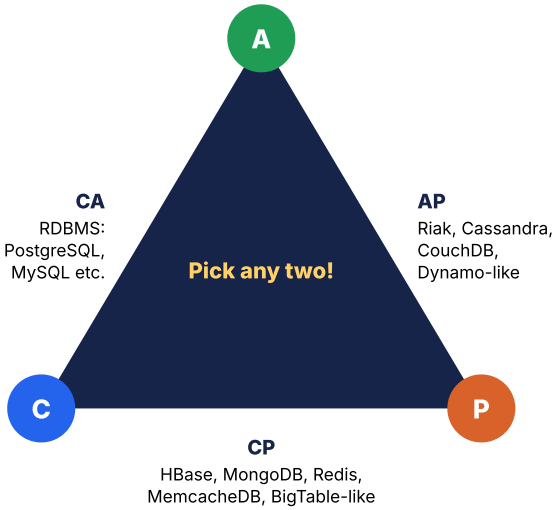


Figure 3.14 — Databases and CAP

Understanding CAP with a story

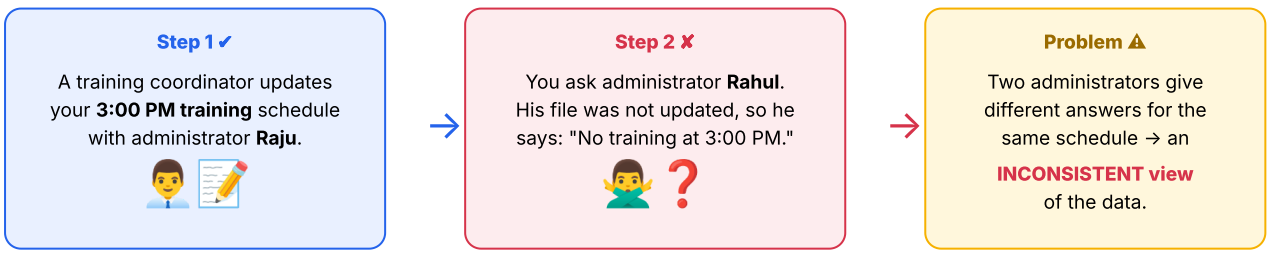


Figure 3.15 — Scenario: consistency

- **Availability:** if an administrator is working, the user should get a response in a reasonable time. Problem: if we force both administrators to **synchronise before every response**, users may have to **wait**.
- **Partition tolerance:** suppose Raju and Rahul **cannot communicate** because of a network problem. A partition-tolerant system **continues to operate** despite this communication break.

C · Choose Consistency during a partition

Stop or delay some operations until nodes agree → correct/latest data, but **some requests may not be served immediately**.

A · Choose Availability during a partition

Keep answering from reachable nodes → requests get responses, but some responses may **temporarily use older data**.

🔑 KEY POINT

Network partitions **will** happen in a distributed system, so P is usually a must. The real choice **during a partition** is Consistency vs Availability.

CAP combinations

Combination	Meaning	Examples (textbook figure)
CA	Consistency + Availability: up-to-date reads and responses while there is **no network partition** to handle	Traditional RDBMS — PostgreSQL, MySQL
CP	Consistency + Partition tolerance: during a partition, prefer **correct/consistent data** even if some requests must wait or fail	HBase, MongoDB, Redis, MemcacheDB, BigTable-like systems
AP	Availability + Partition tolerance: during a partition, keep the system **responsive**, accepting that replicas may **temporarily disagree**	Riak, Cassandra, CouchDB, Dynamo-like systems

PREFER CONSISTENCY WHEN...

- Incorrect or stale data can cause **serious problems**.
- Reads and writes must reflect the **latest confirmed** state.

e.g., **bank balance**, critical inventory update.
Ask: "Can I tolerate an old value?" → No → choose C.

PREFER AVAILABILITY WHEN...

- The service must **keep responding** even during network problems.
- Temporary differences between replicas **can be corrected later**.

e.g., **social feeds**, caches, highly available services.
Ask: "Can I tolerate temporary inconsistency?" → Yes → choose A.

🧠 MEMORY TRICK — WHICH DB IS WHERE?

CA = "**C**lassic **A**pps" → old RDBMS (MySQL, PostgreSQL).
CP = "**C**orrect **P**lease" → **H M R M B**: "**H**appy **M**onkeys **R**un **M**any **B**ananas" — HBase, MongoDB, Redis, MemcacheDB, BigTable.
AP = "**A**lways **P**resent" → **R C C D**: "**R**eal **C**ats **C**hase **D**ogs" — Riak, Cassandra, CouchDB, Dynamo.

Quick revision — terminologies

Term	One-line meaning
In-memory analytics	Keep data in RAM for faster analysis.
In-database processing	Run analytics inside the database to reduce data movement.
SMP	Many processors share common memory and one OS (tightly coupled).
MPP	Many processors/nodes work in parallel with dedicated memory and own OS.
Parallel system	Processors cooperate closely (tightly coupled) on the same work.
Distributed system	Independent networked machines cooperate (loosely coupled).
Shared nothing	Each node owns its processor, memory and disk → fault isolation + scalability.
CAP	Consistency + Availability + Partition tolerance — you can guarantee only two.

✅ **Before you move on — can you…**

☐ Explain the Amazon & truck examples and draw the Data → Information → Knowledge → Insights pyramid?

☐ Draw the "What is BDA" figure and explain its 6 points (incl. moving code to data)? List 6 things BDA isn't?

☐ Explain both schools of classification + the 1.0/2.0/3.0 table and staircase?

☐ List the 7 top challenges and draw the 4-quadrant reactive/proactive figure?

☐ Define data science, draw the Venn diagram, and explain business, technology & maths skills?

☐ Explain the 4 responsibilities of a data scientist with an example?

☐ Explain in-memory, in-database, SMP vs MPP, parallel vs distributed, SM/SD/SN with diagrams?

☐ State CAP, draw the triangle with databases, and say when to prefer C or A?

LAST-NIGHT REVISION

Final Revision

Use these pages the night before the exam

1. Cheat sheet
2. All memory tricks in one place
3. Important 2-mark questions
4. Important 5/10-mark questions

⏱ ≈ 30 minutes | Read the 🔑 boxes again too

★ Cheat sheet

CHAPTER 1 · Digital data

- **Data** = quantities, characters, symbols on which a computer operates; stored/transmitted as electrical signals.
- **Data → Information → Insights.** Data is internal & external; homogeneous & heterogeneous.
- **3 types**: Structured (rows/cols, RDBMS) · Semi-structured (tags, self-describing: XML, JSON, HTML, email) · Unstructured (no model: images, video, chats) = **80–90%**.
- **Cardinality** = no. of rows; **Degree** = no. of columns.
- Constraints: UNIQUE, NOT NULL, business rule, permissible values; PRIMARY KEY; tables related by common column.
- **Ease of structured (ISIST)**: Insert/update/delete (DML) · Security (encryption, tokenization) · Indexing · Scalability (scale up) · Transactions (**ACID**).
- **ACID**: Atomicity (all or nothing) · Consistency (consistent state) · Isolation (as if alone) · Durability (permanent).
- **Semi-structured**: no data model; tags segregate semantics & enforce hierarchy; no data/schema separation; same-class entities may differ in attributes/order. Chars: inconsistent structure, self-describing label/value, schema blended, unknown attributes.
- **XML–SOAP, JSON–REST**; MongoDB & Couchbase/CouchDB store JSON natively.
- **Unstructured**: machine-generated (satellite, atmospheric, radar, CCTV, medical imaging) vs human-generated (emails, office docs, texts, audio, social media). Terminology issues: implied structure; structure not helpful; unanticipated structure.
- **7 techniques**: Data mining (KDD; association rules, regression, collaborative filtering) · Text analytics (categorisation, clustering, sentiment, entity extraction) · NLP · Noisy text analytics · Manual tagging with metadata · POS tagging · UIMA (IBM, open source).
- **Comparison** (7 props): Technology · Transactions · Versioning · Flexibility · Scalability · Robustness · Query performance.

CHAPTER 2 · Big Data

- **3 Cs of data**: Composition (structure, sources, granularity, types, static/streaming) · Condition (usable as-is or needs cleansing?) · Context (where/why generated).
- **Evolution**: ≤1970s mainframes (primitive, structured) → 1980s–90s RDBMS (complex, relational) → 2000s+ WWW & IoT (complex, unstructured; data-driven).
- **Gartner**: "high-volume, high-velocity and high-variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making."
- Data → Information → Actionable intelligence → Better decisions → Enhanced business value.
- **6 challenges**: growth, cloud & virtualisation, retention, lack of skills, capture/storage/processing/security, visualisation. (Fig: capture, storage, curation, search, analysis, transfer, visualisation, privacy.)
- **3 Vs**: Volume (KB→YB, ×1024) · Velocity (batch → periodic → near real time → real time) · Variety (structured, semi, unstructured).
- Sources: internal (file systems, SQL, NoSQL, archives) · external (public web) · both (sensors, logs, social, business apps, media, docs).
- **Other Vs**: Veracity (trust) & Validity (correctness) · Volatility (how long valid) · Variability (inconsistent flow, peaks).

- **Why**: more data → accurate analysis → confidence → efficiency: cost ↓, time ↓, new products, smart decisions.
- **BI vs Big Data**: central server vs distributed FS · vertical vs horizontal scaling · offline vs real-time+offline · structured vs variety · data→code vs code→data · schema-on-write vs schema-on-read.

CHAPTER 3 · Big Data Analytics

- Key idea: large data → analyse → patterns → better decisions (Amazon recommendations, 500-truck example).
- Pyramid: Data (collect) → Information (organise) → Knowledge (summarise, analyse) → Actionable insights (synthesise) → Decision.
- **What BDA is (6)**: technology-enabled · deeper insights · competitive edge · handshake of IT + data scientists + business users · large varied data · moving code to data.
- **Isn't**: only volume · just technology · replace RDBMS · replace DW · only for Google/Amazon · one-size-fits-all RDBMS.
- **Hype**: data growth · cheaper storage · tools available.
- **1st school**: basic · operationalised · advanced · monetised. **2nd school**: 1.0 descriptive (1950s–2009) · 2.0 +predictive (2005–2012, Hadoop) · 3.0 +prescriptive (2012–, in-memory, in-DB, ML). Hindsight → insight → foresight.
- **7 top challenges**: scale · security · schema · continuous availability · consistency · partition tolerance · data quality.
- **Importance — 4 approaches**: reactive BI · reactive BDA · proactive analytics · proactive BDA ★.
- **Data science** = extracting knowledge from data; skills: business acumen + technology + mathematics. Process: collect → prepare → analyse → show → communicate → decide.
- **Data scientist (4)**: data management · analytical techniques · business/domain expertise · communication & presentation.
- **In-memory**: data in RAM. **In-database**: compute inside DB (OLTP → ETL → EDW).
- **SMP**: shared memory, one OS, tightly coupled, caches + system bus. **MPP**: own memory + OS per processor, messaging, harder to program.
- **Parallel** = tightly coupled, user sees one system. **Distributed** = loosely coupled machines over network.
- **SM / SD / SN**; SN advantages: fault isolation + scalability.
- **CAP (Brewer)**: pick two of C, A, P. CA: RDBMS · CP: HBase, MongoDB, Redis, MemcacheDB, BigTable · AP: Riak, Cassandra, CouchDB, Dynamo.

All memory tricks in one place

To remember…	Trick	Expands to
Cardinality vs degree	**C**ardinality = **C**ount of rows	rows = cardinality; columns = degree
5 eases of structured data	**I S I S T**	Insert/update/delete, Security, Indexing, Scalability, Transactions (ACID)
7 techniques for unstructured data	"**D**ark **T**ea **N**ever **N**eeds **M**ore **P**ure **U**pgrades"	Data mining, Text analytics, NLP, Noisy text, Manual tagging, POS tagging, UIMA
7 comparison properties	"**T**he **T**iny **V**ampire **F**lew **S**wiftly, **R**arely **Q**uerying"	Technology, Transaction, Version, Flexibility, Scalability, Robustness, Query
Characteristics of data	The **3 Cs**	Composition, Condition, Context
6 challenges with Big Data	"**G**iant **C**louds **R**arely **L**ack **S**ome **V**iews"	Growth, Cloud, Retention, Lack of skills, Storage/security, Visualisation
3 Vs	A **river**	how much (Volume), how fast (Velocity), how many kinds (Variety)
Data units in order	"**K**ind **M**en **G**ive **T**ea, **P**lease **E**at **Z**esty **Y**ogurt"	KB, MB, GB, TB, PB, EB, ZB, YB (each ×1024)

To remember...	Trick	Expands to
6 points of "What is BDA"	"**T**he **D**eep **C**ompetitive **H**andshake **W**ith **M**ovement"	Technology-enabled, Deeper insights, Competitive edge, Handshake, Working with large data, Moving code
First school of analytics	**BOAM**	Basic, Operationalised, Advanced, Monetised
4 types of analytics	"**D**o **D**octors **P**rescribe **P**roperly?"	Descriptive, Diagnostic, Predictive, Prescriptive
7 top challenges	"**S**ix **S**tudents **S**at **C**almly, **C**opying **P**ast **Q**uizzes"	Scale, Security, Schema, Continuous availability, Consistency, Partition tolerance, Quality
Data scientist's skills	**BTM** three-legged stool	Business acumen, Technology, Mathematics
4 responsibilities	"**M**y **A**unt **B**akes **C**akes"	Manage data, Analyse, Business context, Communicate
SMP vs MPP	"**S**ame **M**emory, **P**lease" / "**M**y **P**rivate **P**artition"	shared memory + one OS / own memory + own OS
CP databases	"**H**appy **M**onkeys **R**un **M**any **B**ananas"	HBase, MongoDB, Redis, MemcacheDB, BigTable
AP databases	"**R**eal **C**ats **C**hase **D**ogs"	Riak, Cassandra, CouchDB, Dynamo

🎯 Important exam questions

Short answers (2–3 marks)

1. Define data. Differentiate quantitative and qualitative data. §1.1
2. What is cardinality and degree of a relation? §1.3
3. Expand and explain ACID. §1.3
4. Why is semi-structured data called self-describing? §1.4
5. Why is an email placed in the unstructured category? §1.8
6. What is UIMA? Who developed it? §1.6
7. What is noisy text? Give two examples. §1.6
8. State the Gartner definition of Big Data. §2.3
9. Differentiate veracity, volatility and variability. §2.6
10. What is schema-on-read? §2.8
11. What is meant by "moving code to data"? §3.2
12. List what Big Data Analytics isn't. §3.2
13. What is in-memory analytics? Advantages? §3.8
14. Differentiate SMP and MPP. §3.8
15. State the CAP theorem. §3.9

Long answers (5–10 marks)

1. Explain the classification of digital data with examples and a neat diagram. §1.2–1.5
2. Explain structured data and the ease of working with it. §1.3
3. Write a note on semi-structured and unstructured data. §1.4–1.5
4. Explain the techniques to find patterns in / interpret unstructured data. §1.6
5. Compare structured, semi-structured and unstructured data. §1.7
6. Explain the evolution and definition of Big Data. §2.2–2.3
7. Explain the 3 Vs of Big Data with sources of data. §2.5
8. What are the challenges with Big Data? §2.4
9. Differentiate traditional BI and Big Data. §2.8
10. What is Big Data Analytics? Explain with a diagram. §3.2
11. Explain Analytics 1.0, 2.0 and 3.0. §3.3
12. Explain the top challenges facing Big Data. §3.4
13. Why is Big Data Analytics important? (4 approaches) §3.5
14. Explain data science and the skill sets of a data scientist. §3.6–3.7
15. Explain shared-nothing architecture and its advantages. §3.8
16. Explain the CAP theorem with an example and the CA/CP/AP databases. §3.9

🎯 HOW TO WRITE A 10-MARK ANSWER

1. **Definition** (use the exact textbook words — the 🔑 / Definition boxes).
2. **Diagram** — neat, labelled, with a figure caption. Draw it early in the answer.
3. **Explanation in points** — one heading per point, 2–3 lines each.
4. **Example** — Amazon, bank, student marks, truck company… examiners love examples.
5. **Conclusion** — 1 line tying back to "better decisions" / "business value".

All the best for CIE-1! 🚀 — Made from your Unit 1 lecture slides (Lectures 1–4) and the textbook by Seema Acharya & Subhashini Chellappan.